

CP 219: MACHINE LEARNING FOR CYBER-PHYSICAL AND MOBILITY SYSTEMS

August-Dec Semester 2026, Project 1 Spec

Competition closes on Sunday, 10:00 AM IST, September 27th, 2026
Report due on Sunday 11:59 PM IST, September 27th 2026

Competition Link: <https://www.kaggle.com/t/2725ee1e58c94ba6b43dfa070c499d25>

Weight: 15% of total mark

Introduction

Language models now produce prose that reads as though a person wrote it. Given a short opening written by a human, a sufficiently large model can continue the passage in a way that is fluent, plausible, and — to a casual reader — indistinguishable from human writing. The models behind this behaviour are trained on very large collections of text drawn from journalism, digitised books, and material scraped from the open web, and they have found their way into products ranging from customer-service agents to translation tools to writing assistants. The same capability creates a problem. When text can be produced at scale and at negligible cost, questions of provenance become hard to answer: who wrote this, and can it be relied upon? Fabricated news, coordinated influence campaigns, and undisclosed automation in settings where authorship matters all become cheaper to run. One response is to build classifiers that examine a passage of text and estimate whether a person or a machine produced it.

Your Task

You are given a corpus of documents, each written either by a human or generated by a language model, and your job is to build a classifier that tells the two apart. The training data contains 10,536 documents, each with an id (row identifier), the document encoded as a sequence of token identifiers, and a class label. The test data consists of 3,000 documents, for which you are given the id and the token sequence but not the label. *Your task is to predict, for every test document, whether it was written by a human or generated by a machine.*

Every document has been tokenised and encoded before release: each integer identifies one token in a fixed vocabulary, and the mapping itself is not distributed, so the original words cannot be recovered. You are working with token statistics and sequence structure alone. Documents vary in length, from a handful of tokens to a little under two thousand; any model expecting fixed-width input will require you to decide how to handle this.

To make the project fun, we run it as an in-class competition on Kaggle, one of the most popular platforms for predictive modelling and analytics, where you will compete against other students in the class. The following sections give more details on data format, the use of Kaggle, and the marking scheme. Your assessment will be based on your final ranking in the competition, the absolute score that you achieve, and your report. The marking scheme is designed so that you will pass if you put in effort. So fear not and embrace the power of machine learning!

Data Format

You will have access to three types of files, primarily `train_data`, `test_data`, and `sample_submission.csv` file. **These files are available in a Zip file in Class Team under “Project 1 Data and Specification” Channel in “File” section.** Each of the two data files is provided in two interchangeable formats: a JSON Lines version (`.json`, one JSON object per line, with the token sequence already parsed as a list of integers).

train.json — contains 10,536 labelled documents drawn from a vocabulary of token identifiers in $\{0, \dots, 18437\}$. Each entry includes:

- **id**: a unique identifier for each data point (row) in the training data. Identifiers are arbitrary and carry no information about the class.
- **text**: the document, represented as a sequence of integer token identifiers. Token 0 is reserved for rare words excluded from the vocabulary.
- **label**: the class of the document, encoded as A or B, where A denotes a human-written document and B a machine-generated one.

test.json — contains 3,000 documents. Each entry includes the same fields as the training set, with the label omitted:

- **id**: a unique identifier for each data point (row) in the test data
- **text**: the document, represented as a sequence of integer token identifiers, encoded exactly as in the training data and drawn from the same vocabulary.

A `sample_submission.csv` file shared with you shows an example submission file. The `sample_submission.csv` file contains the same ids in its first column as those found in the `test_data` file, listed in the same order. Once you have done predictions for each id in the `test.json` file using your own ML model, you should create a submission file in CSV format, similar to the `sample_submission.csv` (described next), and upload this to Kaggle.

If your prediction is B for the first test document, A for the second test document, then your output file should be as follows in CSV format:

```
id,label
21994204,B
81419419,A
...
```

The first line should be a header, exactly as shown in `sample_submission.csv`. There should be 3,000 rows (excluding the header row) in total, each with a unique id. The second column in the `sample_submission.csv` corresponds to the predictions (which are given as a constant placeholder). The IDs of predictions should match the IDs in the `test_data` file, in the same order. The simplest approach is to load `sample_submission.csv` and overwrite its second column.

The test set will be used to generate the classification accuracy for your performance. During the competition, the classification accuracy on a subset of the test set will be used to rank you in the leaderboard; the remaining portion is held back and determines your final ranking, which is released only after the competition closes. As we provide no explicit validation set, you may want to reserve part of the training partition for this purpose during model development. Your job is to develop an algorithm that can automatically capture the nuances of the problem, in order to generalise well to unseen data (estimated here over the test set).

Note that you will have to provide your code in your submission (discussed later) and we will run it at our end to ensure you have not done any cheating. The use of external datasets is strictly prohibited. You may only work with the data provided. Do not attempt to locate the original data sources, as we have applied several obfuscation measures to prevent this.

Kaggle In-class Competition

You can join this competition only using the following link. Please note that you won't be able to participate in the competition otherwise.

Link: <https://www.kaggle.com/t/2725ee1e58c94ba6b43dfa070c499d25>

Please do the following by the end of the first week (07/09/2026) after receiving this assignment:

- Setup an account on Kaggle with username and email being your IISc student email.
- Form your team of student peers (Note that some or all teams may be formed by the Course Instructor to make sure each team has a student with some prior experience with programming)
- Connect with your team mates on Kaggle as a Kaggle team, using a team name that you chosen in the Google form shared in the class. You can choose any team name. Only submit your entries via the team; and
- Register your final team using the Google Form. Link: <https://forms.gle/3JCVjgfo4v2TLMwUA>). There are three sections in this form. First section asks your basic details. The second section is for those who have already decided/ registered a team on Kaggle competition. The last section is for those who are looking for a team member(s) to form a team.

Teams should consist of three individuals. If you cannot find a team, please introduce yourself to fellow students in workshop or the lecture, or post to *Project 1 Discussion Channel* in Class Team that you are in search of a team. Feel free to mention your skills/strength and what you are looking for in other members. In only very rare occasions, we will permit teams of less than three (and we will mark all teams based on our expectations of what a team of three could achieve). If you still cannot find a team, fill out the Google form shared above by EOD on 01/09/2026. Based on the responses, we will form the teams. The motivation for working in teams is that in industry, practicing machine learning experts work effectively in teams.

The real labels for the test data are hidden from you, but were made available to Kaggle. Each time a submission is made, the predictions in 50% of the test data will be used to calculate your public score and determine your position in the public leaderboard. This information will become available from the competition page almost immediately. At the same time, the remaining 50% predictions are used to compute a private accuracy and rank in the private leaderboard, and this information will be hidden from you. At the end of the competition, only private scores and private ranks will be used for assessment and will be revealed publicly. This type of scoring is a common practice and was introduced to discourage overfitting to public leaderboard. A good model should generalize and work well on new data, which in this case is represented by the portion of data with the hidden accuracy.

The evaluation score used in this competition is the classification accuracy over all classes, defined as the number of instances labelled correctly as a fraction of the total number of instances. Each participant can do a maximum of 3 submissions every day. Before the end of the competition, each team will need to choose 2 best submissions for final scoring. These do not have to be the latest submissions. By default, your top two submissions based on public leaderboard score are chosen, then after competition close these two submissions will be scored on the entire test set, and the best total test scoring submission will make up your accuracy. If you prefer non-default submissions for your two “scored private submissions” Kaggle permits you to select others prior to completion of the competition.

Remark: Note that teams will be limited to 3 submissions per day. You must not submit using your individual account or create multiple accounts to bypass the daily submission limit. **You should only make submissions using the team name, individual submissions are not allowed and may attract penalties (-3 marks).**

Report

Each team (only one member per team) must submit a report that describes, analyzes, and, where applicable, compares the methods used. There is no fixed template, but the report should include the following:

1. The Kaggle team name, names of all team members, and the last five digits of each student ID. You may also include your final ranking on the private leaderboard.
2. A brief description of the problem and an introduction to any notation used in the report.

3. A concise account of data preprocessing, validation, sampling strategies, and class distributions you explored. Mention any noteworthy aspects of your software setup or experimental pipeline, challenges you faced, and what you learned. These issues are often as important as the choice of learning algorithm.
4. A description of your final approach(es), including the motivation and reasoning behind them, and an explanation of why you believe they performed well or poorly in the competition.
5. A discussion of alternative methods you considered and why you selected your final approach. Reflect on the strengths and weaknesses of the methods you tried. If you experimented with different models or hyperparameters, provide comparisons supported by reasoning—not just raw performance numbers (e.g., avoid explanations limited to “method A achieved 0.60 accuracy, method B achieved 0.70, so we used method B”).
6. If you applied feature transformations, feature selection, or feature engineering, describe them, explain your expected outcomes, and report the observed effects after evaluation. When comparing methods, consider using evaluation metrics beyond accuracy to better capture model behavior, especially for rare or minority classes.

When describing the algorithms, aim for clarity and conciseness. Write at a level that a postgraduate student can easily follow. If you are using established algorithms, provide a brief summary that demonstrates your understanding, along with appropriate references, rather than reproducing the full description. In the report, we are particularly interested in seeing your reasoning and decision-making process in selecting one algorithm over another.

The report should be submitted as a PDF, and be no more than three A4 pages of content, including all plots, tables and references¹ (single column, font size of 11 or more and margins at least 1 cm, much like this document). You do not need to include a cover page. **If a report is longer than three pages in length, we will only read and assess the report up to page three and ignore further pages..** Marks may be deducted if these instructions are not followed. Learning to communicate ideas concisely in short reports is a vital skill valued in industry, government, and academia alike. We highly recommend using LATEX for writing your report.

Submission and Assessment

In summary, **each team (only one student per team)** is required to make the following submissions for this project:

- **By 27/09/2026, 10AM:** One or more submission files with predictions for test data (at Kaggle Website). This submission must be of the expected format as described above, and produce a place somewhere on the leaderboard. Invalid submissions do not attract marks for the competition portion of grading;
- **By 27/09/2026, 11:59PM:** Report in PDF format (via "Assignment" Section for this project in Class Teams);
- **By 27/09/2026, 11:59PM:** Source code used in this project as a single ZIP archive (via "Assignment" Section for this project in Class Teams). Your code can be in any of the following languages C, C++, Python, Jupyter Notebook, R or MATLAB. If there is another language you like to use, please contact us first. If the language requires compiling, a makefile or script must be provide to build the executables. We may or may not run your code, but we will definitely read. You must include a README.txt file that briefly explains the purpose of each file. **Please DO NOT include the training or test data file in the ZIP file.**
- **By 27/09/2026, 11:59PM:** Each group must hold a meeting every week of the project and record it using the **Weekly Meeting Minutes** template available under "Project 1 Discussion Channel" on Teams. The minutes record what each member completed that week and plan for the next week. They attract no marks, but all

¹Plots can be useful for model selection, assessing convergence, features importance, displaying results and model interpretation, among other things. For instance, plotting the parameters of your model with respect to the objective function can often give insights into what the model has learned.

weekly minutes are submitted as a single appendix PDF with your final report, and a group that cannot produce them has no record to draw on if contributions are later disputed.

After the project we will run an unmarked group evaluation quiz, where you give staff feedback on your own contribution and your teammates'. If it identifies a significant dispute we reserve the right to follow up — drawing on your minutes — and apply differential marking. This is rare as almost all teams work well together.

The project will be marked out of 30. No late submission of Kaggle portion will be accepted; late submissions of reports will incur a deduction of 3 marks per day, or part thereof. Based on our experimentation with the project task and the design of the marking scheme below, we expect that all reasonable efforts at the project will achieve a passing grade or higher. So relax and have fun!

Marking Scheme

Kaggle competition (15 marks) This mark takes into account both achieved accuracy, as well as your standing in the class. Assuming N is the number of students, and R is your rank in the class, the mark you get for the competition part is

$$12 \times \frac{\max\{\min(acc, 0.97) - 0.32, 0\}}{0.65} + 3 \times \frac{N - R}{N - 1}$$

The first term constitutes up to 12 marks, and rewards high accuracy systems with a maximum score for excellent systems with $\geq 97\%$ accuracy, and zero score to those with scores $\leq 65\%$. The second term, worth 3 marks, is based on your rank and is designed to encourage competition and innovation. Ties are handled so that you are not penalised by the tie. All who are tied will get the same marks for score, but ranking will be decided based on total number of submission entries. The score with fewer entries will be ranked higher among tied ones.

Report (15 marks) The report will be marked using the rubric in Table 1.

Bonus Mark (1 mark) One bonus mark is available to groups whose report shows a genuine contribution of their own, beyond applying the methods covered in class. Examples include an innovative feature or representation you devised, an error analysis that led you to change your approach, or a well-executed negative result. To be awarded, the claim must rest on a controlled comparison: the same validation protocol, with and without the contribution, and the numbers reported. The mark is not competitive — every group meeting the bar receives it — and it is not awarded for using a larger or more advanced model architecture. You need to provide this information in your report with proper justification, to get this 1 bonus mark.

Plagiarism policy

You are reminded that all submitted project work in this subject is to be your own individual work. Automated similarity checking software (Turnitin) will be used to compare submissions. It is University policy that cheating by students in any form is not permitted, and that work submitted for assessment purposes must be the independent work of the student(s) concerned. For more details, please see the policy at <https://iisc.ac.in/about/student-corner/academic-integrity/>.

Critical Analysis (8 marks)	Report Clarity and Structure (7 marks)
7–8 <i>marks</i> Final approach is well motivated and its advantages/disadvantages clearly discussed; thorough and insightful analysis of why the final approach works/not work for provided training data; insightful discussion and analysis of other approaches and why they were not used	6–7 <i>marks</i> Very clear and accessible description of all that has been done, a postgraduate student can pick up the report and read with no difficulty
5–6 <i>marks</i> Final approach is reasonably motivated and its advantages/disadvantages somewhat discussed; good analysis of why the final approach works/not work for provided training data; some discussion and analysis of other approaches and why they were not used	4–5 <i>marks</i> Clear description for the most part, with some minor deficiencies/loose ends (e.g., there are no- table gaps and/or unclear sections)
3–4 <i>marks</i> Final approach is somewhat motivated, and its dvantages/disadvantages are discussed; limited analysis of why the final approach works/not work for provided training data; limited discussion and analysis of other approaches and why they were not used	2–3 <i>marks</i> Generally clear description, but there are notable gaps and/or unclear sections.
1–2 <i>marks</i> Final approach is barely or not motivated and its advantages/disadvantages are not discussed; no analysis of why the final approach works/not work for provided training data; little or no discussion and analysis of other approaches and why they were not used	1 <i>mark</i> The report is unclear on the whole, omits all key reference, and the reader can barely discern what has been done

Table 1: Report marking rubric.